



2 Channels AI Dashcam

DS100

Specification



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Preface

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Product Name Explanation

DS100 is the dual-lens model with one built-in road-facing camera and one built-in cabin-facing camera.

Symbol Explanation

The following symbols are used to categorize and explain potential hazards that may arise from improper operation

Symbol	Explanation
	Warning: Conditions that could jeopardize the safety of the user or cause injury
	Important: Conditions that may lead to data corruption or damage to the device's firmware or hardware
	Note: Additional information, explanations of terms, and other relevant details

Abbreviation Explanation

Acronym	Full Name
ADAS	Advanced Driver Assistance System
DSC	Driver Safety Cockpit
DMS	Driver Monitoring System
LDW	Lane Departure Warning
HMW	Headway Monitoring Warning
FCW	Forward Collision Warning
PCW	Pedestrian Collision Warning

Product Overview

Product Introduction

The DS100 is an AI-powered dashcam designed to enhance driver safety and improve fleet management efficiency. Leveraging advanced AI technology, it actively detects risky driving events and unsafe behaviors, providing real-time alerts to drivers to help avoid potential hazards. The system also uploads event data to the fleet management platform for driver training purposes. Additionally, it transmits real-time vehicle location and operational data to the platform, offering high-quality remote communication and video playback to streamline fleet management and boost overall efficiency.

Product Features

- Built-in forward high-definition road camera supporting 1080P video recording.
- Built-in in-cabin ultra-wide-angle lens supporting up to 1080P video recording.
- Supports H.264/H.265 encoding.
- Supports up to 1*1TB external Micro SD card storage, and Built-in 1*eMMC storage (Up to 256GB) allowing simultaneous storage of both main and sub-streams.
- Built-in 4G communication module (eSIM supported), and GNSS positioning module
- Audio and video data supports AES-256 encryption, data transmission uses encrypted TLS 1.3 protocol.
- Supports 2-channel input, 1-channel output, 1-channel CAN and 1-channel RS232.
- Compact design, does not obstruct the driver's view on both large and small vehicles.
- Supports OBD power supply, convenient and quick installation.
- Built-in ADAS features (3.85 mm road-facing lens), supports lane departure warning, forward collision warning, pedestrian collision warning and headway monitoring warning.
- Built-in DSC function, supports detection of poor driving behavior.
- Supports echo suppression algorithm to improve two-way intercom quality.
- Supports sleep mode and remote wake-up.
- Built-in 6-axis gravity sensor, supports detection of rapid acceleration, rapid deceleration, sharp turns, and collisions.
- Built-in WIFI/Bluetooth, supports Bluetooth IoT peripherals such as fuel level sensor, tire pressure sensor, speed sensor, etc.
- Supports 2 sets of button functions: one set of emergency alarm buttons and one set of function buttons.

AI Features

The DS100 uses advanced video analysis technology and machine vision to automatically detect road hazards and unsafe driver behaviors. The system continuously monitors and triggers both visual and auditory alerts to remind the driver to pay attention to safety. Additionally, relevant alarm footage is automatically uploaded to the cloud for real-time monitoring and review.

➤ ADAS Features Diagram

			
LDW	HMW	FCW	PCW

DSC Features Diagram

			
Lens Covering	Yawning	Phone	Smoking
			
Distraction	No Driver	Seat belt	

Specification

General	
System	Embedded Linux
Language	Options: Traditional Chinese, English, Spanish (Latin American), Spanish (European), Portuguese (Latin American) Default: English * The language includes both interface language and voice reminders. TTS supports only Chinese and English
Video & Audio	
Video & Audio Recording	2-channel video + 1-channel audio
Max. Capability	Road Facing: 1920*1080P@25fps Driver Facing: 1920*1080P@20fps
Image Setup	Adjustable brightness, chroma, contrast, color saturation, and sharpness
Video Coding	Options: H.264 and H.265 Default: H.265
Audio Compression Standard	Options: ADPCM, G.711, and G.726. Default: ADPCM
Encoding Standard	Options: VBR and CBR. Default: VBR
Microphone	Built-in
Loudspeaker	Built-in 3W loudspeaker, with adjustable volume, not less than 70 dB at 1 m distance
Parameters of road facing lens	
Sensor Type	1/2.9" 2-megapixel CMOS sensor
Shutter Speed	1/30s~1/100000s
Lens	- Version 1: Focal length: 3.85 mm FOV (H) = $86.4 \pm 3^\circ$ FOV (V) = $46 \pm 3^\circ$ FOV(D) = $103 \pm 5^\circ$ - Version 2: Focal length: 2.58 mm FOV (H) = $124 \pm 3^\circ$ FOV (V) = $67.5 \pm 3^\circ$ FOV(D) = $147 \pm 5^\circ$
Minimum Illuminance	Color: 0.05 lux/F1.2
Lens Mount	Built-in lens
Wide Dynamic Range (WDR)	Digital WDR
Signal-to-Noise Ratio (S/N)	$\geq 48\text{dB}$
Parameters of driver facing lens	

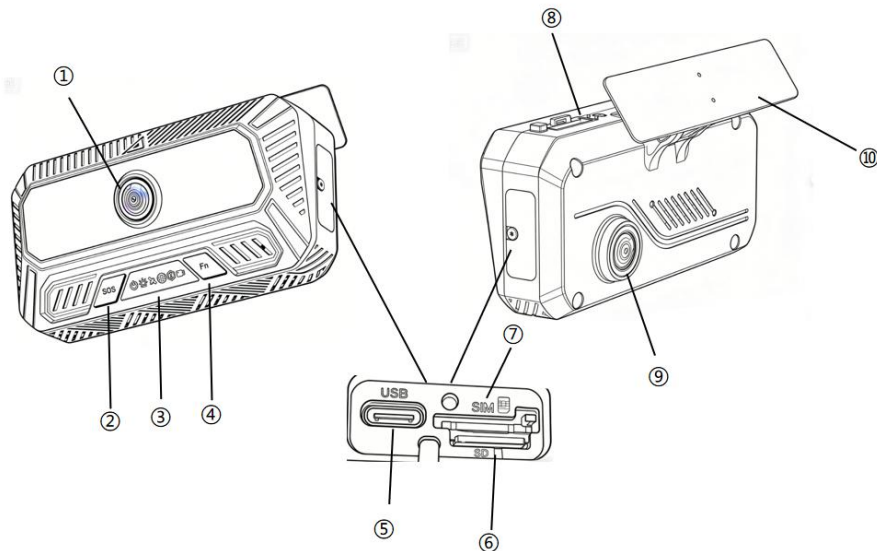
Sensor Type	1/2.9" 2-megapixel CMOS sensor																														
Shutter Speed	1/30s~1/100000s																														
Lens	Focal length: 2.2 mm FOV (H) =146.6±3° FOV (V) =80.2±3° FOV(D)=172±5°																														
Lens Mount	Built-in lens																														
Wide Dynamic Range (WDR)	Digital WDR																														
Backlight Compensation	Supported																														
Signal-to-Noise Ratio (S/N)	≥45dB																														
Infrared	Supported. The built-in environmental light sensor turns on/off the infrared automatically * Threshold: 4 lux from daytime to night, and 8 lux from night to daytime. There may be slight variations depending on the device. Please refer to actual measurements																														
LED Indicator Status																															
<table><tr><th>Indicator Light</th><th>Icon</th><th>Off</th><th>On / Flashing</th></tr><tr><td>Power Indicator</td><td></td><td>No power supply</td><td> Power supply normal</td></tr><tr><td>Alarm Indicator</td><td></td><td>No alarm currently</td><td> (Flashing) Alarm triggered</td></tr><tr><td>GPS Signal Indicator</td><td></td><td>GPS positioning normal</td><td> Positioning error  (Flashing) Poor positioning quality</td></tr><tr><td>Network Status Indicator</td><td></td><td>Connected to server</td><td> Not connected to server  (Flashing) Airplane mode</td></tr><tr><td>WiFi Status Indicator</td><td></td><td>WiFi is in Disable or Client mode</td><td> AP mode  WiFi error</td></tr><tr><td>Recording Status Indicator</td><td></td><td>Normal recording</td><td> Recording stopped / Malfunction</td></tr></table>				Indicator Light	Icon	Off	On / Flashing	Power Indicator		No power supply	 Power supply normal	Alarm Indicator		No alarm currently	 (Flashing) Alarm triggered	GPS Signal Indicator		GPS positioning normal	 Positioning error  (Flashing) Poor positioning quality	Network Status Indicator		Connected to server	 Not connected to server  (Flashing) Airplane mode	WiFi Status Indicator		WiFi is in Disable or Client mode	 AP mode  WiFi error	Recording Status Indicator		Normal recording	 Recording stopped / Malfunction
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Storage																															
Micro SD card	Micro SD card×1, (SDXC 128GB/256GB/512GB/1TB) Read/write rate: Class 10 or above is recommended																														
Built-in Storage	Built-in eMMC storage(optional), Up to 256GB																														
Sensor																															
Six-axis Sensor	Harsh acceleration, Harsh deceleration, Harsh cornering, and accident detection																														
Environmental Light Sensor	Supported, used as the cockpit camera, subject to day/night switching																														
Port																															
RS232	1-channel																														
I/O Port	2-channel input, 1-channel output																														

CAN	1-channel (standard J1939 protocol)
USB	1 × Type C USB2.0
Button	2*Button, One is dedicated to SOS, and the other is a function button. To switch Wi-Fi to AP mode, press the button twice within 2 seconds * For more details on the use of this button, please refer to the product's user manual
Network	
Wi-Fi/BT	Support 2.4G (IEEE Std.802.11a/IEEE Std.802.11b/IEEE Std.802.11g /IEEE Std.802.11n) Support BT 5.2, Supports Bluetooth IoT peripherals
4G	Plug-in SIM card (Nano SIM card), reserved eSIM patch position, can support pre-burn-in eSIM. Plug-in SIM card and eSIM cannot be used simultaneously. ● For Europe and Asia: LTE FDD: B1/B3/B5/B7/B8/B20/B28 LTE TDD: B38/B40/B41 ● For Latin America: LTE FDD: B2/B3/B4/B5/B7/B8/B28/B66 ● For North American* LTE FDD: B2/B4/B5/B12/B13/B66  An industrial SIM card (MP2) is required. The use of a standard SIM card (MP1) is prohibited. We are not responsible for any issues caused by the use of a standard SIM card Note: *Under development.
Positioning	
GNSS	Supported GPS L1 C/A:1575.42 MHz GLONASS L1: 1598.0625 - 1605.375 MHz Galileo E1:1575.42 MHz BDS B1I: 1561.098 MHz QZSS L1 C/A:1575.42 MHz
Power related	
Power supply	12V and 24V vehicles (self-adaptive)
Power consumption	● In standby mode: about 0.072W ● In sleep mode (4G, GNSS, and MCU powered): about 0.456W ● Typical power consumption (with single Storage installed and SIM card for dialing, 2-channel video, and infrared lamp turned on): about 4W ● Full-load power consumption (with dual Storage cards installed, SIM card for dialing, Wi-Fi turned on, 2-channel video and infrared lamp turned on): about 4.3W

	* The above data are test data obtained in a specific environment in the laboratory, and may vary with the individual product differences, service environment, and testing methods
Environment	
Operating Temperature	-40°C ~ +70°C (-40°F ~ +158°F)
Storage Temperature	-40°C ~ +85°C (-40°F ~ +185°F)
Operating Humidity	15~95% non-condensing
Storage Humidity	15~95% non-condensing
IP Rating	IP30 (not waterproof)
Dimensions and Weight	
Dimensions L×W×H	126.68*62.47*48.56mm
Weight	about 0.5kg
* The actual dimensions and weight may vary with the individual product differences, manufacturing processes, and testing methods.	
Package Contents	
DS100 ×1, standard power cable ×1, 2-meter extension cable × 1, Cable fixing clip× 1, hex wrench ×1, mounting bracket ×1, bracket bolt ×1, desiccant ×1, alcohol wipe ×1	
* Contents may vary depending on the region and specific requirements	

Product Appearance and Component Names

Front & Side View

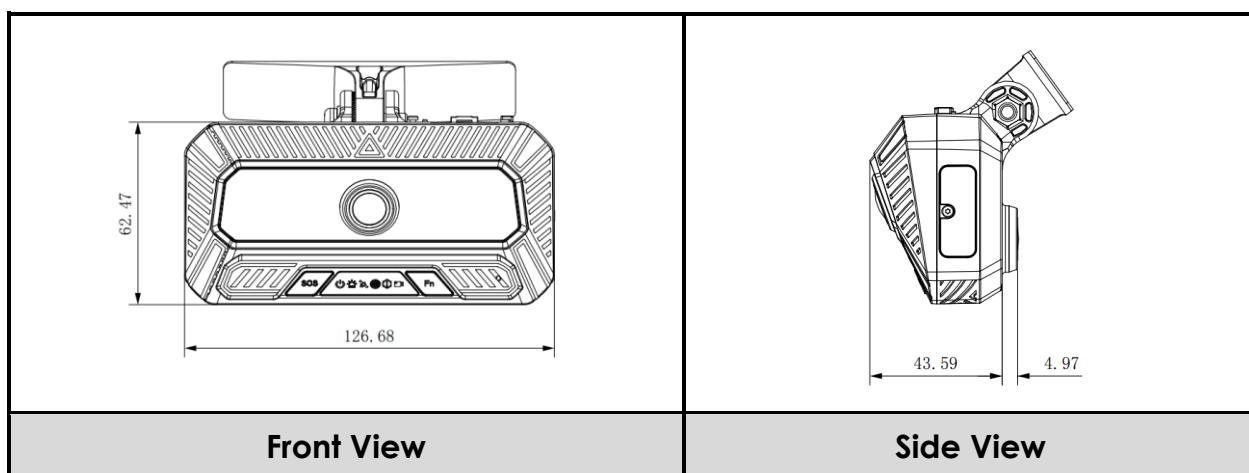


No.	Components	No.	Components
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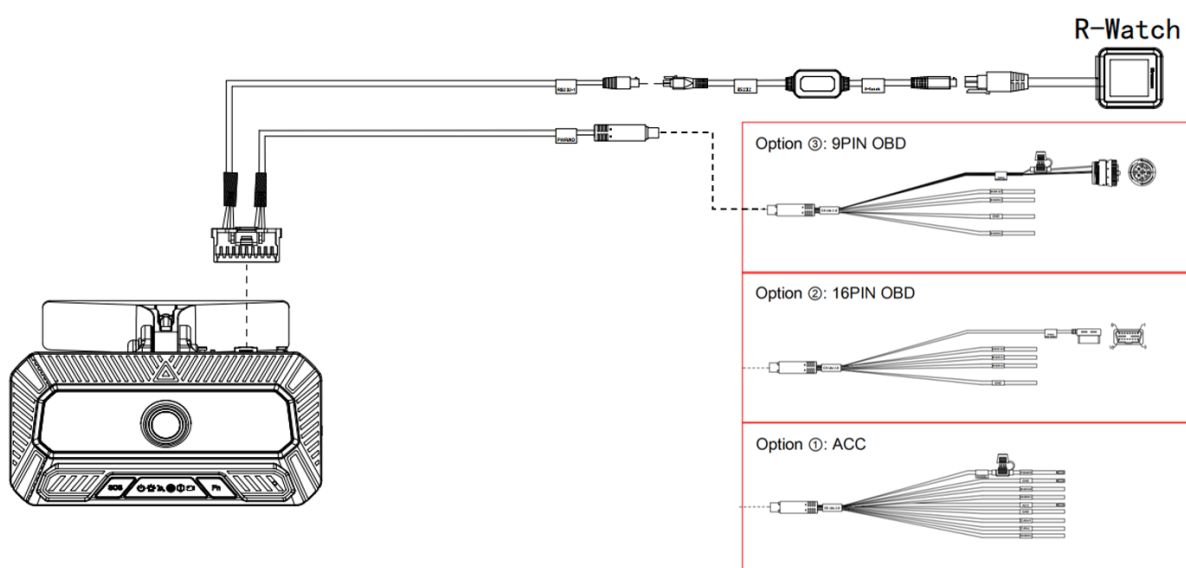
1	Cabin-facing Camera	6	Micro SD Card Slot
2	Emergency Alarm Button	7	SIM Card Slot
3	Status Indicator Light	8	Cable fixing clip
4	Function Button	9	Road-facing Camera
5	USB Type C Port	10	Installation Bracket

Dimension Diagram

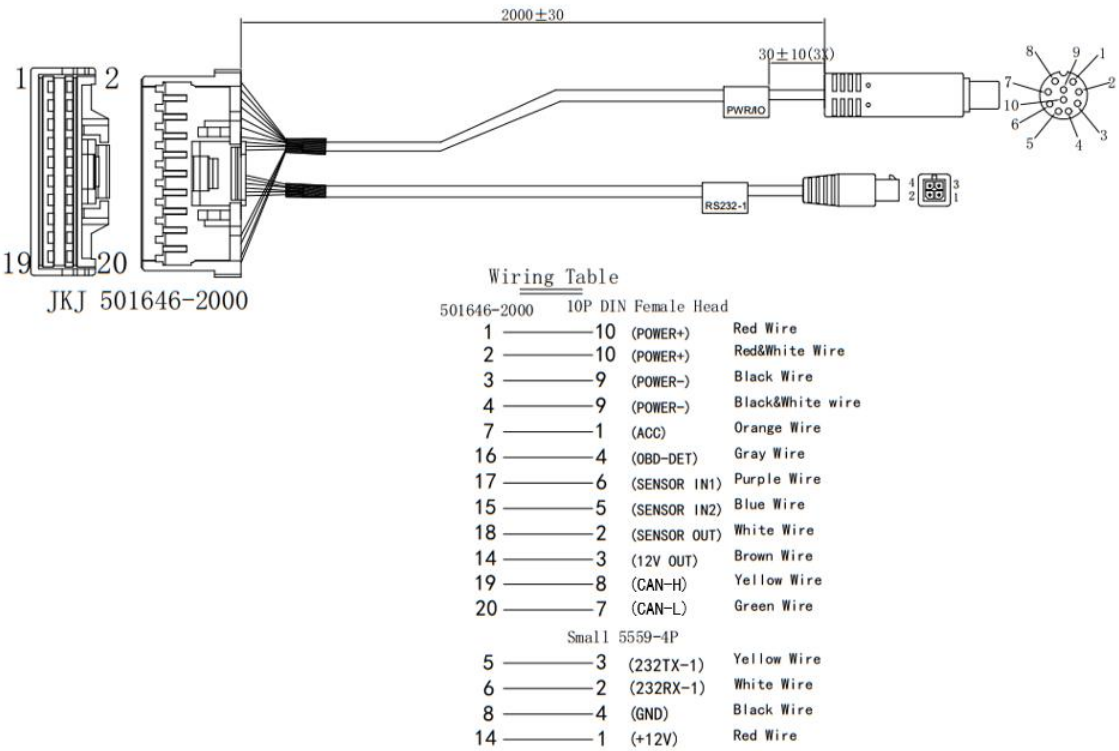
(Unit: mm)



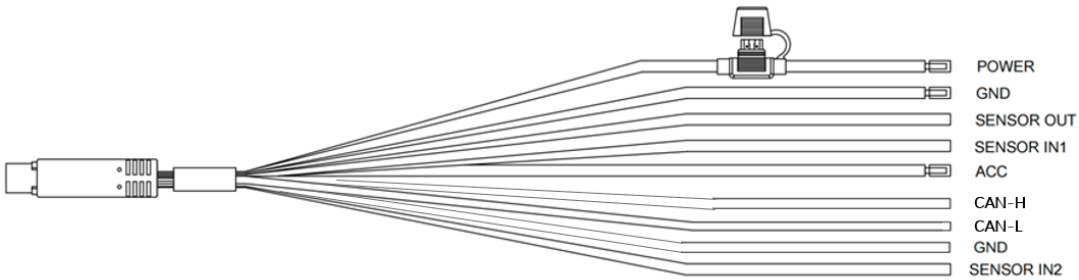
System Connection Diagram



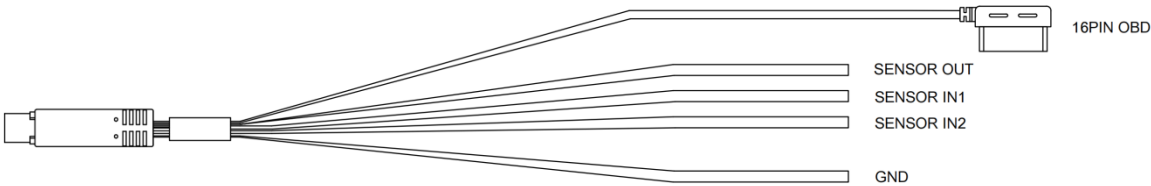
Cable Details



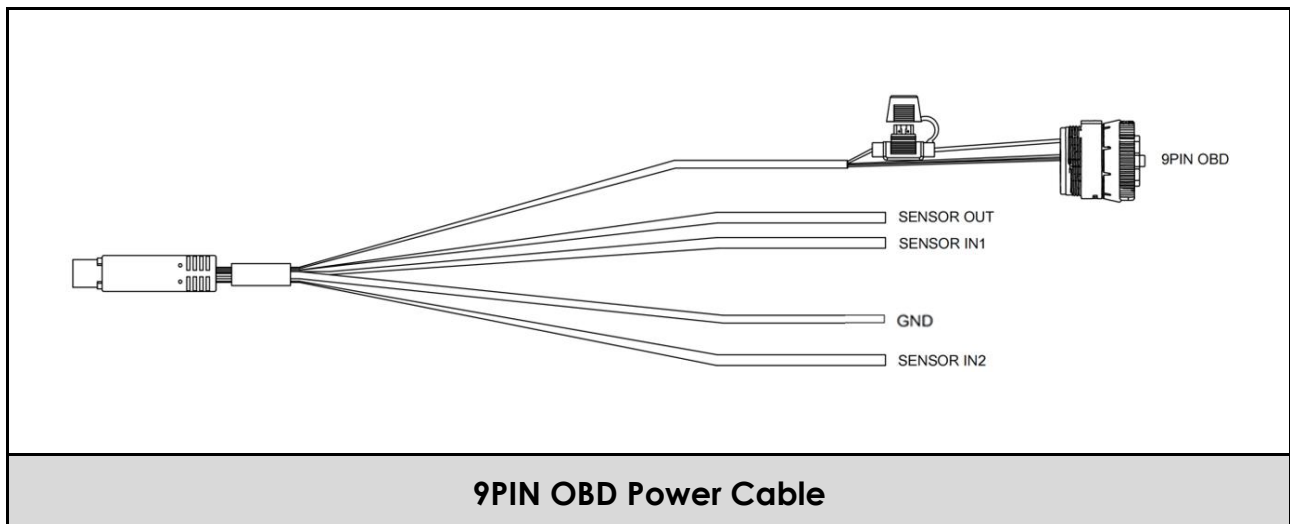
Power Cable



ACC Power Cable



16PIN OBD Power Cable — Optional



Special Instructions

- This product requires installation by professionals; otherwise, there is a risk of electric shock, damage to the vehicle wiring, impact on AI performance, and device detachment.
- When used under direct sunlight, the surface temperature of this product may exceed 60°C. Please avoid touching the sun-exposed surface to prevent burns.